

Standalone Sign Language Converter Arm

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Abstract

This project presents a **Standalone wearable gesture-to-voice translator glove** developed for assistive communication.

The system uses **flex sensors** to detect finger bending and an Arduino Nano to process **gesture inputs in real time**.

Recognized gestures are mapped to **predefined voice messages**, which are played through a speaker using an audio module.

Unlike camera-based systems that rely on OpenCV, this design eliminates the need for external devices and complex image processing.

The glove is **compact, portable, and energy-efficient**, making it suitable for **daily use**.

Special focus is given to selecting simple, unique, and reliable gestures for accurate detection.

Additional features such as **customizable gestures, multilingual output, and emergency signals** can be integrated.

Overall, the project provides a **low-cost** embedded solution for **real-time communication assistance**.

LITERATURE SURVEY

Reference	Publication Title	Core Advantage	Primary Limitation
[1]	Sign Gesture Recognition (IEEE, 2020)	High accuracy for complex dynamic gestures.	Computational heavy; requires cameras and ideal lighting.
[2]	Real-time SLR (IEEE, 2022)	Bidirectional communication (Sign-to-Speech/Text).	Susceptible to background noise and motion blur.
[3]	ASL-English AI Framework (MDPI, 2023)	Robust AI translation for full sign languages.	Dependency on cloud processing and internet.
[4]	Data Glove System (2019)	High precision for finger-specific bending data.	High hardware cost; limited to hand/finger movement.
Novelty	Our Project	Standalone, low-cost, and energy-efficient.	Predefined gesture library only.

AIM & OBJECTIVES

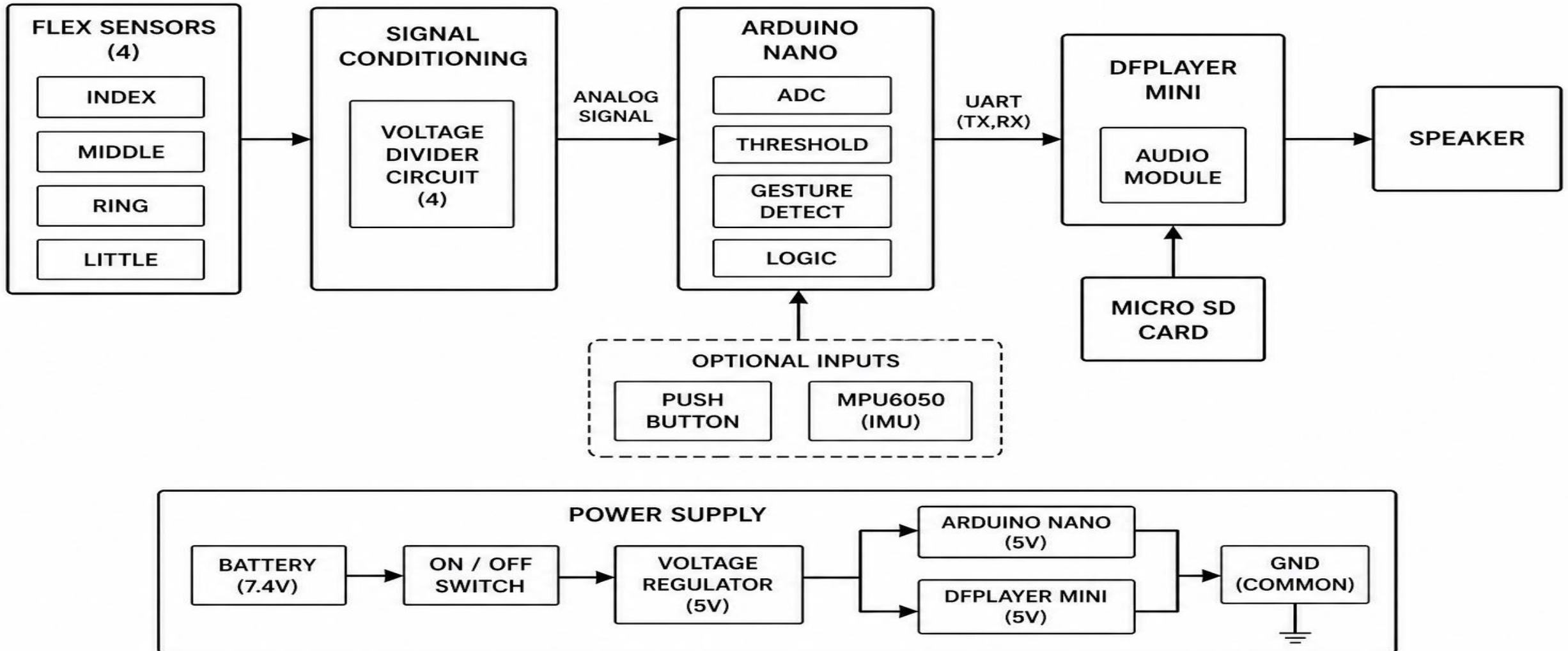
Aim

To design and develop a standalone wearable glove that converts hand gestures into voice output for assistive communication.

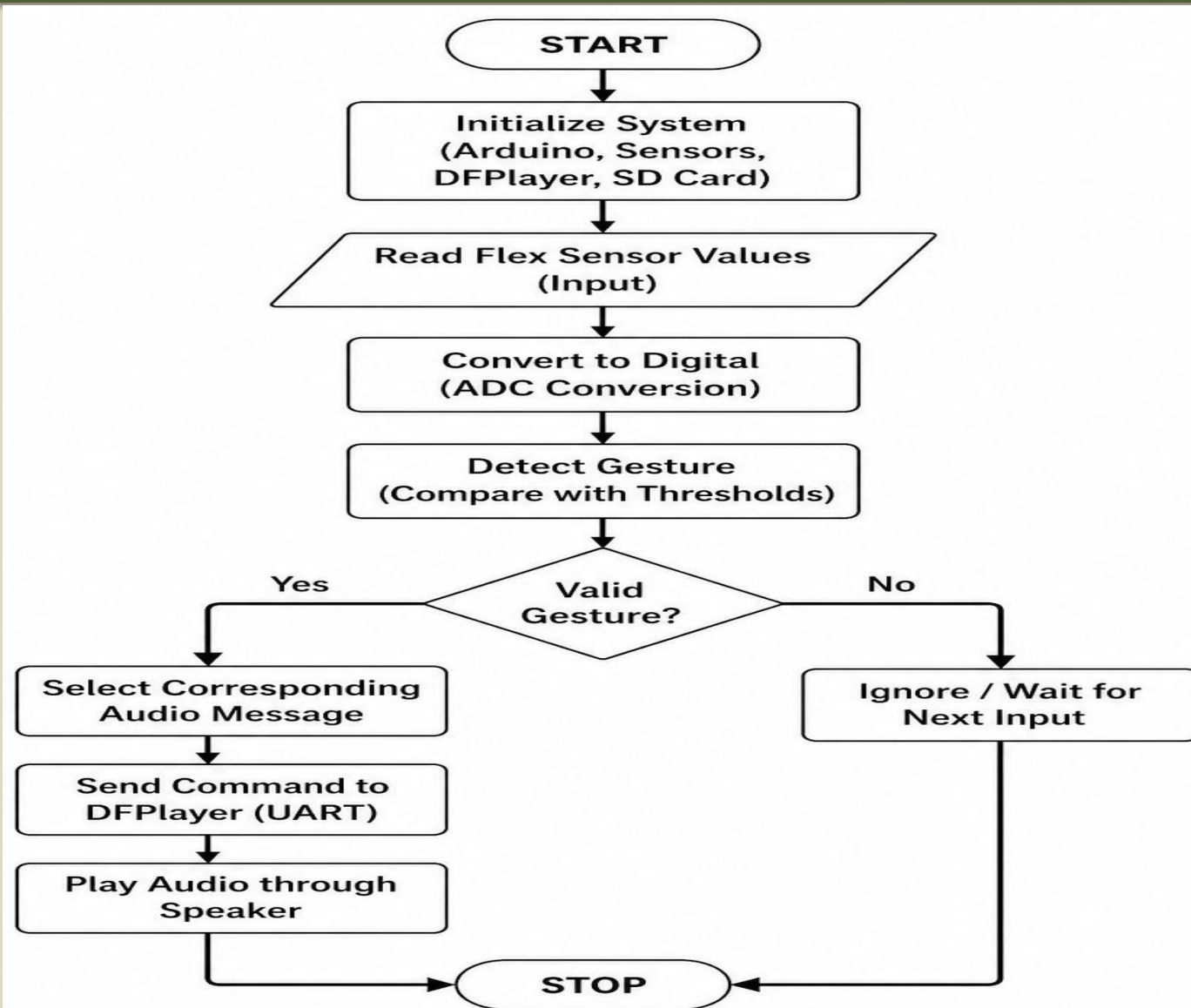
Objectives

- To detect finger movements using flex sensors
- To convert sensor data into digital signals using Arduino Nano
- To identify predefined gestures using embedded logic
- To generate corresponding voice output using DF-Player module
- To ensure the system is portable, low-cost, and easy to use
- To design simple and reliable gesture patterns for accurate detection

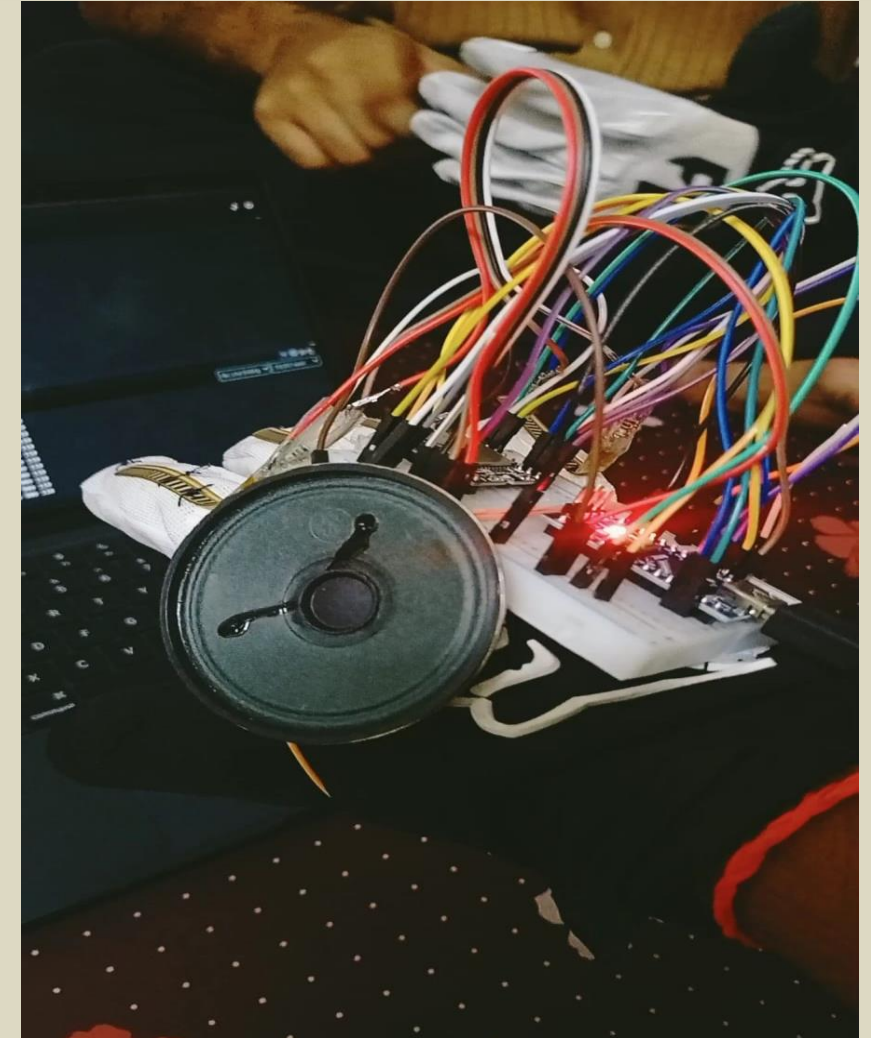
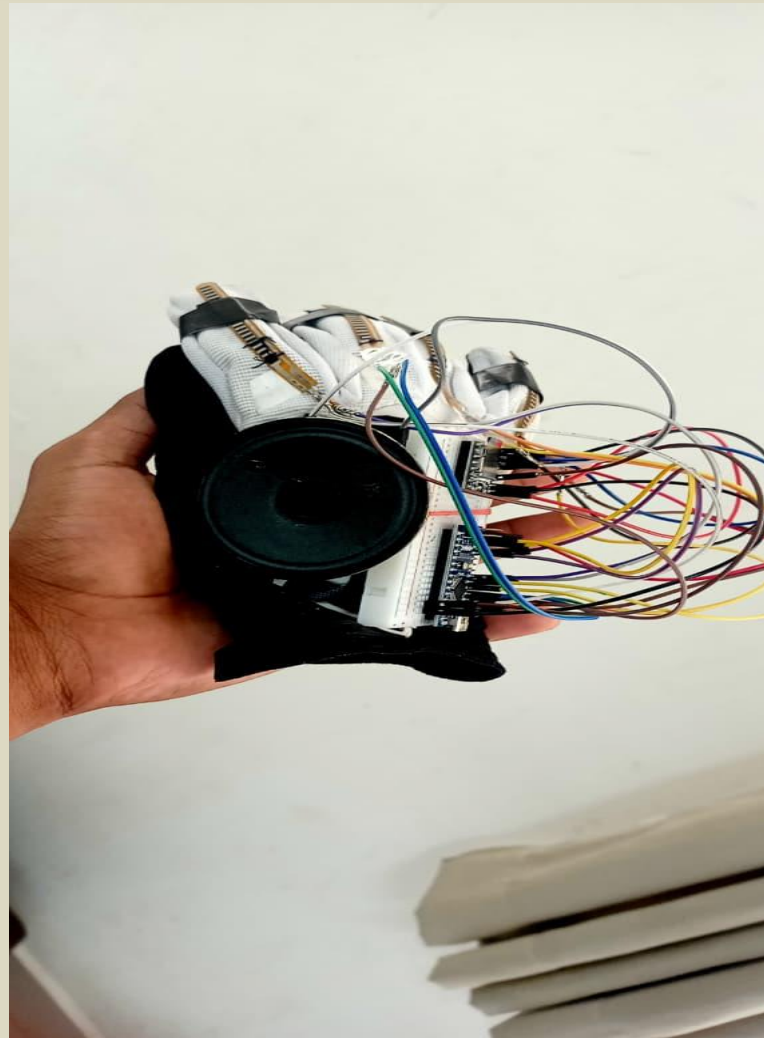
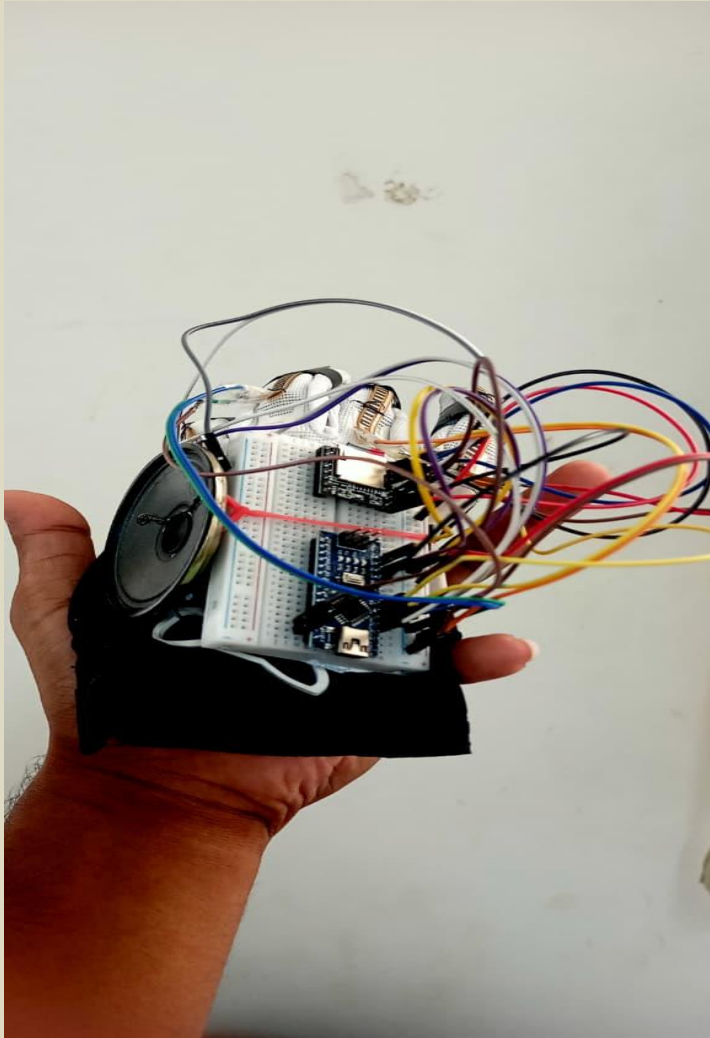
Block Diagram



Flow chart of the Program



Picture of the Project



Hyperlink of the project video: <https://www.youtube.com/shorts/OMqGuuNs2ac>

Budget in detail(with components)

Component	Purpose	Cost (INR)
Arduino Nano	The main controller	₹230
Flex Sensors (x4)	2.2" sensors for finger bend	₹1,120
DF Player Mini	MP3 Audio Module	₹100
Micro SD Card (8GB)	Voice message storage	₹ 300
3W 8-Ohm Speaker	Audio output	₹50
Battery + Battery holder	Portable power source	₹75
Glove + Hardware	Mounting, wires, resistors, velcro	₹605
Total Cost		₹2,480

The project successfully demonstrates a standalone wearable gesture-to-voice translator glove for assistive communication. Using flex sensors and an Arduino Nano, the system detects finger movements, processes them, and converts them into voice output through an audio module. Unlike camera-based systems, it is compact, low-cost, and does not require external devices, making it suitable for real-time use. The optimized gesture design ensures reliable performance. Overall, the project proves an effective and practical embedded solution for communication assistance, with scope for future improvements like additional gestures and smart features.

Outcome of the project

- Developed a working prototype of a gesture-to-voice glove
- Achieved accurate gesture detection using flex sensors
- Successfully generated real-time voice output for gestures
- Designed a low-cost and portable embedded system
- Improved understanding of sensor interfacing and embedded programming
- Demonstrated practical application in assistive communication

- [1] P. S. Neethu, J. S. Suganthi, and S. R. S. Rose, **"Real-time Sign Language Recognition using Machine Learning and Neural Network,"** *2022 6th International Conference on Computing Methodologies and Communication (ICCMC)*, Erode, India, 2022, pp. 605-611. [View on Google Scholar](#)
- [2] M. A. Ahmed, A. S. Albahri, et al., **"Real-time sign language framework based on wearable device: analysis of MSL, DataGlove, and gesture recognition,"** *Soft Computing*, vol. 25, pp. 11101–11122, May 2021. [View on Google Scholar](#)
- [3] V. D. Avina, M. Amiruzzaman, S. Amiruzzaman, L. B. Ngo, and M. A. A. Dewan, **"An AI-Based Framework for Translating American Sign Language to English and Vice Versa,"** *Information*, vol. 14, no. 10, p. 569, Oct. 2023. [View on Google Scholar](#)
- [4] S. S. Rao et al., **"Deep Learning based Hand Gesture Recognition for Sign Language Translation,"** *2024 International Conference on Computer, Information and Communication Technology (ICC/CT)*, Mumbai, India, 2024, pp. 1-6. [View on Google Scholar](#)